Faculty: Computers and Informatics **Class Year:** 2021/2022 **Year:** First Year **Subject Name:** Operations Research	**Mid-term Exam**	**Date: april.** 2022 **Time Allowed:** 1 Hour. **No. of Pages: 1** **No. of Questions: 2** **Model : 2**

Answer the following questions: **(15 points)**

Question 1:Mark (A) for true and (B) for false:

1. The term sensitivity analysis refers to testing how a problem solution reacts to changes in one or more of the model parameters.
2. A model is a mathematical representation of a problem situation including variables, parameters, and equations
3. Decision variables limit the degree to which the objective in a linear programming problem is satisfied. — constraints
4. Any linear programming problem can be solved using the graphical solution procedure — only with 2 desicion variables
5. Fixed cost is the difference between total cost and total variable cost.
6. The set of solution points that satisfies all of a linear programming problem's constraints simultaneously is defined as the feasible region in graphical linear programming.
7. In a linear program, the constraints must be linear, but the objective function may be nonlinear
8. Any time that we have an isoprofit line that is parallel to a constraint, we have the possibility of multiple solutions.
9. Once management scientist makes his or her decision and recommendation to management, then typically, his or her involvement with the problem is finished.
10. The second step of the management science process is to define the problem.
11. If variable costs increase, but price and fixed costs are held constant, the break even point will increase.
12. Unbounded solution in an LPP is where the objective function can be increased or decreased — increased only
finitely.
13. In linear programming, constraints can be represented by inequalities only. — inequalities and equalities
14. In simplex method we add artificial variables to ≥ constraint to satisfy non negativity constraint.
15. The difference between the left-hand side and right-hand side of a less-than-or-equal-to constraint is referred to as slack.
16. The break even point is the volume that equates total revenue with total cost.
17. It is possible To have exactly two optimal solution to a linear programming problem.
18. A linear programming problem can be both unbounded and infeasible.
19. The break even point help managers to ensure profit.
20. The mathematical theory behind linear programming states that an optimal solution to any problem will lie at a(n) corner point or extreme point of the feasible region.

Question 2: choose the correct answer:

solve the following problem using simplex method then answer the following.

$$max\ z = 13x_1 + 11x_2$$

sub to:

$$4x_1 + 5x_2 \leq 1500$$
$$5x_1 + 3x_2 \leq 1575$$
$$x_1 + 2x_2 \leq 420$$
$$x_1, x_2 \geq 0$$

21. The value of s_2 is .....
a) 1575 b) 0 c) 240 d) 105

22. The value of s_3 is .....
a) 315 b) 0 c) 420 d) 105

23. The value of z is .....
a) 4095 b) 2700 c) 2400 d) 4335

24. The value of x_1 is .....
a) 315 b) 270 c) 240 d) 0

25. The value of x_2 is .....
a) 0 b) 270 c) 240 d) 75

26. The value of s_1 is .....
a) 1500 b) 45 c) 240 d) 105

27. Is x_1 , x_2 basic variable.
a) true b) false

28. Is s_2 basic variable.
a) true b) false

29. Is s_1 basic variable.
a) true b) false

30. Is s_3 basic variable.
a) true b) false

Best wishes

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